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An Adaptive Two-Scale Image Fusion of Visible and Infrared Images

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ABSTRACT In this paper, we proposed an adaptive two-scale image fusion method using latent low-rank representation (LatLRR). First, both IR and VI images are decomposed into a two-scale representation using LatLRR to generate low-rank parts (the global structure) and saliency parts (the local structure). The algorithm denoises at the same time. Then, the guided filter is used in the saliency parts to make full use of the spatial consistency, which reduces artifacts effectively. With respect to the fusion rule of the low-rank parts, we construct adaptive weights by adopting fusion global-local-topology particle swarm optimization (FGLT-PSO) to obtain more useful information from the source images. Finally, the resulting image is reconstructed by adding the fused low-rank part and the fused saliency part. The experimental results validate that the proposed method outperforms several representative image fusion algorithms on publicly available datasets for infrared and visible image fusion in terms of subjective visual effect and objective assessment.

INDEX TERMS Fusion global-local-topology particle swarm optimization (FGLT-PSO), guided filter, image fusion, latent low-rank representation (LatLRR).

I. INTRODUCTION

Image fusion is designed to combine multiple input images from different modal sensors into a single image. The fused image is expected to provide more comprehensive information for the human visual system or further applications as compared to any of the input images. Du *et al.* [1] give a general survey on fusion methods, especially in the field of medicine. Infrared and visible image fusion, which has already been widely applied in many fields, such as surveillance and military [2], has always been an important problem in the image fusion field. Li *et al.* [3] presented a comprehensive research on the application of image fusion and meanwhile proposed that a good fusion method has the following properties: first, it can preserve most of the useful information of different source images; second, it does not produce artifacts; and third, it is robust to imperfect conditions like mis-registration and noise.

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Various image fusion methods [1]–[4] have been proposed. Among these methods, multiscale transform image fusion methods [3] are the most popular tools used in various image fusion scenarios. Classical multiscale transform based fusion methods include pyramid-based methods like the Laplacian pyramid (LP) [5], wavelet-based methods like the dual-tree complex wavelet transform (DTCWT) [6], and multiscale geometric analysis based on ones like the non-subsampled contourlet transform (NSCT) [7]. Besides, in some methods, classical filters are used iteratively to get the multiscale transform. For instance, Zhang *et al.* [8] used a Gaussian filter to do the decomposition and used the co-occurrence filter to fuse the base layer and detail layer in a different way. This method has outstanding performance in retaining small details. In general, the multiscale transform based fusion methods consist of the following three steps [3]: (1) conducting multi-scale decomposition on the source images; (2) fusing transform coefficients at each scale with fusion strategy; (3) reconstructing the fused image with a relevant inverse transform. Although theses multiscale transform

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based methods have excellent performance in image fusion as they are consistent with the human visual system, there are still some drawbacks. First, the fused images may have some distortions and artifacts because the spatial consistency is not well-considered in these methods [9]. In addition, in some multiscale geometric analysis based methods, there are adverse impacts of noise and mis-registration from the source images on the fused result, due to the down-sampling operations in the decomposition and reconstruction process, which can introduce the pseudo-Gibbs phenomenon. Guided filter based methods [9], generalized random walks [10], and Markov random fields [11] are common methods to make full use of the spatial context and reduce the artifacts mentioned above. The guided filter based method is the most efficient and most extensively used method to reduce the artifacts compared with the other two methods. Besides, the guided filter based method may alleviate the impact of the mis-registration to some extent. The second drawback is that all these conventional fusion multiscale methods require some priori knowledge of the original images [12]. Further, most of the methods are complex and time-consuming with the increase of the decomposition levels and it is difficult to select the decomposition level. In addition, some fusion methods based on optimization have also become popular, such as fusion in the spectral total variation domain [13] and the fusion methods with weighted least square optimization [14].

Compressive sensing (CS) [12], [15], [16] has become another popular transform tool for image fusion in the past decade. This type of method represents information signals in a sparse and compressible way by using a small number of non-adaptive linear projections to reintegrate the image information. Sparse representation (SR) [17] is the most representative algorithm among the CS-based methods. It has superiority in reducing storage space and computation cost, and simplicity on the hardware side [16]. Zhu *et al.* [17] proposed a SR based method combining the advantages of the multiscale method, which achieved satisfactory performance. What is more, the whole representation process can be completed without assuming any a priori information of the source images, which will make the fusion process more efficient than the traditional multiscale transform based methods. Latent low-rank representation (LatLRR) [18], as the improved version of the low-rank representation (LRR) [19] has attracted a large extent of interest in the CS community in recent years, because it can extract salient local features by choosing a more compact dictionary without supervision. Moreover, it is robust to noise and outliers due to the low rank constraint condition. Li et al. use the LatLRR to fuse the infrared and visible images [20]. However, there are still several artifacts in the edge areas due to the lack of spatial consistency in their method. Furthermore, the simple fusion rules (averaging the global parts and simply adding the saliency parts) in their method have as result that it cannot obtain the ideal information from the source images in the fusion process. The same situation also occurs in [17]. Cheng *et al.* [21] use LatLRR for constructing the weight map for the base layer after a shearlet decomposition, which is time-consuming because the construction and decomposition process are not parallel. Hence, the selection of the fusion rules is crucial, especially for the infrared and visible image fusion. The two types of images have very different characteristics: their contributions are different with various illumination conditions and weather conditions. Choosing a fusion rule that can adapt to different conditions is a vital aspect in image fusion.

We therefore propose an adaptive two-scale image fusion scheme based on guided filter and Fusion Global-Local-Topology Particle Swarm Optimization (FGLT-PSO) in this paper. We take advantage of the LatLRR to get the low-rank parts and the saliency parts. We choose adaptive weights using FGLT-PSO [22] as the fusion rule of the low-rank part, rather than a simply average fusion strategy. Moreover, we use the guided filter to construct the saliency weights to reduce the artifacts in our method. The proposed algorithm thus makes the following contributions:

(1) Unlike most of the multiscale transform based methods, the LatLRR is utilized to decompose the source images, which is more efficient and robust to noise and outliers.
(2) Adopting FGLT-PSO to construct the adaptive weights in the low-rank part, the algorithm ensures fusion performance stability and robustness on different images.
(3) With the combination of LatLRR and guided filter, our method has outstanding performance in removing noise and reducing image distortion, which can omit the denoising process of image preprocessing to some extent.
(4) We introduce a scoring mechanism called rank score, which can comprehensively and intuitively evaluate the performance of algorithms taking all the metrics into consideration. Our method is superior to the other methods in terms of adaptivity and robustness in different types of source images according to the rank score scheme.
(5) Only setting several parameters and objective functions will make a good trade-off between efficiency and quality, which illustrates that our method is simple and easy to operate.

The remainder of this paper is organized as follows: relevant technical background is briefly reviewed in Section 2. In Section 3, the proposed fusion method is introduced in detail. The experimental results and analysis are shown in Section 4. The conclusions are presented in Section 5.

II. TECHNICAL BACKGROUND

A. LATENT LOW-RANK REPRESENTATION

As a compressed sensing technique, low-rank representation (LRR) was first proposed by Liu *et al.* in 2010 [19]. Unlike the well-known sparse representation (SR), LRR aims at finding the lowest-rank representation of a collection of vectors jointly, which benefits the acquisition of the global structure of data and is robust against noise and outliners.

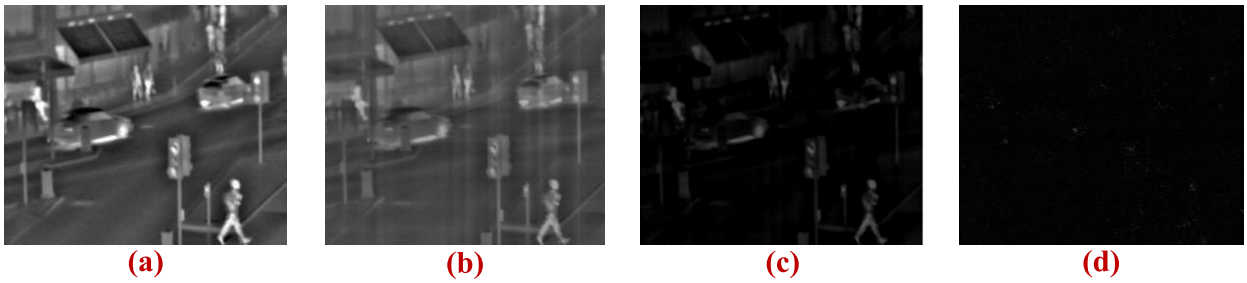


FIGURE 1. The LatLRR decomposed operation. (a) Original image; (b) approximate part; (c) salient part; (d) sparse noise.

However, the LRR method cannot preserve the local structure, namely the salient features. To solve this problem, the authors later presented the latent low-rank representation (LatLRR) [18].

The key aspect of the LatLRR is that an image matrix can be decomposed into global structure (low-rank representation), local structure (salient features), and sparse noise via choosing a more compact dictionary, i.e., both observed and unobserved data in the image matrix itself as the dictionary. The key point is interpreted to solve the following optimization problem according to [18]:

$$\min_{Z,L,E} \|Z\|_* + \|L\|_* + \lambda \|E\|_1 \tag{1}$$

$$s.t.,\ X = XZ + LX + E, \tag{2}$$

where $\lambda > 0$ is the balance coefficient, $\|\cdot\|_*$ denotes the nuclear norm and $\|\cdot\|_1$ is the l_1-norm. $X \in R^{M*N}$ is the image matrix, Z the low-rank coefficient, L the saliency coefficient and E the sparse noise. Then the low-rank representation part XZ, the saliency part LX and the sparse noisy part E can be derived. The noise has some negative effects on the fusion process [3]. Many algorithms are dedicated to denoising before the fusion process. In the low rank representation, the noisy part can be separated from the source images due to the property of the low-rank representation, which helps to omit the extra noise removal process to some degree. An example of the decomposition of the source image using Eq. (1) is show in Figure 1. Figure 1(a) is the original image; Figure 1(b) shows the approximate parts of the images; Figure 1(c) depicts the salient features and Figure 1(d) presents the sparse noise. We can see that the images are well-decomposed into these three parts by using the LatLRR method.

B. GUIDED FILTER

The guided filter is an edge preserving smoothing filter that was first proposed by He *et al.* [9]. It can make full use of correlations between adjacent pixels, so that the image can contain special consistency. The computation time of the guided filter is independent of the filter size, which makes it efficient. The guided filter is based on a local linear model using a guidance image Y as one of the inputs to filter the input image X [9]. As a result, the output image Z can preserve the major information of the input image and acquire the variation tendency of the guidance image simultaneously. Assuming that the filtering output Z is a linear transformation of the guidance image Y in a local window v_k centered at a pixel k, it can thus be described as

$$Z_i = a_k Y_i + b_k \quad \forall i \in v_k \tag{3}$$

where Z_i and Y_i is the ith pixel value of the output and guidance images, v_k is a window of size $(2r+1) \times (2r+1)$. The linear coefficients a_k and b_k are the constant in the window v_k and these constants can be estimated by minimizing the squared difference between the output image Z and input image X as

$$E(a_k, b_k) = \sum_{i \in v_k} \left((a_k Y_i + b_k - X_i)^2 + \eta a_k^2 \right) \tag{4}$$

where η is a regularization parameter deciding the blur degree of the guided filter. The coefficients a_k and b_k can directly be estimated using

$$a_k = \frac{\frac{1}{|v|} \sum_{i \in v_k} Y_i X_i - \mu_k \bar{X}_k}{\delta_k^2 + \eta} \tag{5}$$

$$b_k = \bar{X}_k - a_k \mu_k \tag{6}$$

In this paper, we use the following equation to represent the guided filtering operation:

$$Z = G(X, Y, \eta, r) \tag{7}$$

where G represents the guided filter operator, and r is the radius of the window which decides the filter size.

C. FUSION GLOBAL-LOCAL-TOPOLOGY PARTICLE SWARM OPTIMIZATION

Particle swarm optimization (PSO), originally designed and introduced by Kennedy and Eberhard [23], is a population-based stochastic search algorithm. The algorithm finds the optimal solution that satisfies the objective function during the entire iteration through the cognition and information sharing among the individuals in the swarm. Due to the simple concept and easiness to determine the parameters, PSO has aroused great interest in recent years. Nevertheless, PSO is easily trapped in local extrema and cannot handle complex multimodal problems. To solve this problem, we choose the fusion global-local-topology particle

swarm optimization (FGLT-PSO) [22] from numbers of variant algorithms, because it can jump out of the local optima promptly while maintaining the advantages of the original PSO algorithm.

The FGLT-PSO algorithm starts with choosing a population of randomly generated particles in d-dimensions as the initial solution. Every particle is shown as a vector $\left(\vec{X}_i, \vec{V}_i, \vec{P}_i\right)$ where $\vec{X}_i = (x_i^1, x_i^2 \cdots x_i^d)$ and $\vec{V}_i = (v_i^1, v_i^2 \cdots v_i^d)$ are the position and velocity respectively, and $\vec{P}_i = (p_i^1, p_i^2 \cdots p_i^d)$ is the personal best position (P_{best}). In addition, the best position among all particles is represented as:

$$\overrightarrow{P_g} = \left(p_g^1, p_g^2 \cdots p_g^d\right) \tag{8}$$

During the iteration process, $\vec{P}_g$ is separated into the $\vec{P}_{lbest} = (p_{lbest}^1, p_{lbest}^2 \cdots p_{lbest}^d)$ (the best position found by its neighbors) and the $\vec{P}_{gbest} = (p_{gbest}^1, p_{gbest}^2 \cdots p_{gbest}^d)$ (the best position found by the swarm). Then using $\vec{P}_{lbest}$ and $\vec{P}_{gbest}$ to update the velocities and positions of the ith particles respectively:

$$\begin{aligned} v_i^d(t+1) &= \omega(t) \times v_i^d(t) + C_1(t) \times rand_1 \\ &\quad \times \left(p_i^d(t) - x_i^d(t)\right) + C_2(t) \times rand_2 \\ &\quad \times \left(p_{lbest}^d(t) - x_i^d(t)\right) \end{aligned} \tag{9}$$

$$\begin{aligned} x_i^d(t+1) &= x_i^d(t) + v_i^d(t+1) + C_3(t) \times rand_3 \\ &\quad \times \left(p_{gbest}^d(t) - x_i^d(t)\right) \end{aligned} \tag{10}$$

In Eqs. (9) and (10),$rand_1$, $rand_2$ and $rand_3$ are random numbers in the interval [0,1] and the variable t stands for the iteration step. ω is the inertia weight which reflects the tradeoff capacity between the global and local exploitation and is computed as

$$\omega(t) = \omega_{\max} - \frac{t \times (\omega_{\max} - \omega_{\min})}{T} \tag{11}$$

$C_1(t)$, $C_2(t)$ and $C_3(t)$ are the acceleration coefficients and can be written as:

$$C_j(t) = C_{j\min} + \frac{t \times \left(C_{j\max} - C_{j\min}\right)}{T}, \quad j = 1, 2, 3 \tag{12}$$

where t and T are the current iteration and the total iterations.

During the $(t+1)^{\text{st}}$ iteration, the fitness value of each particle is evaluated according to the objective function as follows:

$$f\left(x_i^d(t+1)\right) = p\left(y_i^d(t+1) \middle| x_i^d(t+1)\right) \tag{13}$$

where $y_i^d(t+1)$ is the observation information of the particle $x_i^d(t+1)$. The iteration procedure can be terminated when the convergence criterions (e.g. a maximum number of computation iterations or the threshold of the objective function) are satisfied.

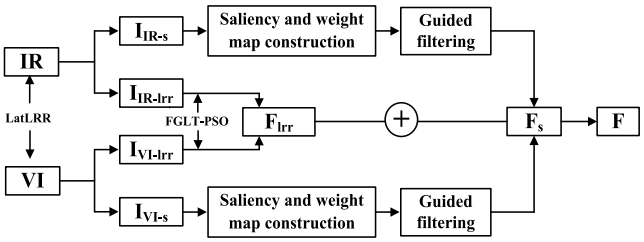


FIGURE 2. Schematic visualization of our proposed method.

III. PROPOSED FUSION METHOD

A. FUSION FRAMEWORK

In this section, we present the proposed adaptive two-scale image fusion algorithm for visible and infrared images. Figure 2 shows the flow chart of the proposed method, which consists a number of essential stages:

1. The source visible image and infrared image are represented by I_{VI} and I_{IR} respectively. Two scale sub-images including low-rank parts (I_{VI_lrr}, I_{IR_lrr}) and saliency parts (I_{VI_s}, I_{IR_s}) for each source image are obtained by LatLRR
2. The saliency parts are fused by the average gradient strategy and the fused saliency image is shown as F_s. At this part, we added the guided filtering method to reduce the artifacts.
3. The low-rank parts of the final fused image are obtained by an adaptive weighted method, in which the weights are optimally estimated using the Fusion Global-Local-Topology Particle Swarm Optimization algorithm. The fused low-rank part is represented as F_{lrr}.
4. The fused image F is reconstructed by simply adding the fused low-rank part (F_{lrr}) and the saliency part (F_s).

In the following subsections, we provide more details of our image fusion process.

B. FUSION OF LOW-RANK SUB-IMAGES

In this section, the adaptive weight fusion rule based on the FGLT-PSO is described. The information in the low-rank parts is vital for the natural appearance of an image, because they contain the bulk of global structure information, brightness information and energy. Weight averaging and arithmetic combination [24] based methods are simple and widely used in the image fusion. However, the source images are obtained under different circumstances (different types of weather and illumination) by multimode sensors, which leads to different contributions of the infrared and visible images to the final fused images according to the features of the source images respectively. The weight averaging based approach treats the infrared and visible images as equivalent parts in the fusion process, which will recede the robustness and cannot guarantee stable performance when handle different types of images. Here, adaptive weights are obtained to assign a larger weight to the part contributing more, by optimizing multiple objective functions using FGLT-PSO. With the introduction of the adaptive weights, the algorithm ensures the fusion

performance to be stable and the robust on different images. The details of our algorithm are as follows:

Step1: Initialize the FGLT-PSO with arbitrary weights α_i. These initial weights are of size M×N, the same size as I_{VI_lrr} and I_{IR_lrr}.

Step2: The adaptive weighted image fusion is carried out as:

$$F_{lrr}(i,j) = \alpha_i \times I_{VI_lrr}(i,j) + (1-\alpha_i) \times I_{IR_lrr}(i,j) \quad (14)$$

Step3: Apply the reconstruction process to obtain the fused image F.

Step4: Evaluate the fused image with multiple objective functions to check whether the optimum of the functions or the maximum number of iterations is reached. If any of these conditions is satisfied, then save the present weights or keep updating the weights according to Eqs. (9) and (10) and repeat Steps2-4.

We summarize these steps in Algorithm1.

Algorithm 1 Fusion of Low-Rank Sub-images

Input: Infrared sub-image I_{IR_lrr}, visible sub-image I_{VI_lrr}, parameters ω, C, *rand*
Output: α_N
1. Set a population of random particles (total number: N)
2. **for** i = 1: Maximum iteration **do**
3. **if** meet the condition that the result less than the threshold of the objective function
4. Update V by Eq. (9)
5. Update X by Eq. (10)
6. Update ω by Eq. (11)
7. Update C by Eq. (12)
8. Calculate the result by Eq. (21)
9. **End if**
10. **End for**

We choose edge retentiveness ($Q^{AB/F}$) [24] and standard deviation (SD) [25] as the objective functions, since one single evaluation metric cannot measure the quality of an image comprehensively. Finally, we choose the Pareto optimal solution [26] as the final weights. In addition, some parameters setting details are shown in section 4.A.

C. FUSION OF SALIENCY SUB-IMAGES

The saliency part contains the local structure, viz. saliency features, special brightness distribution parts, etc. At this part, we introduce the guided filter to suppress the artifacts appeared in the fused image and eliminate the effect of unregistered images on the fusion process to some extent. Moreover, the traditional maximum absolute value is used as the fusion rule in the normal fusion method, but it results in a significant loss of the complementary information, which makes the saliency part not so clear. By contrast, the average gradient strategy (AVG) we use in this part can transfer more salient scene information from each source image into the fused saliency part.

The gradient map is constructed as given in Eq. (15)

$$G_n = \frac{1}{M \times N} \sum_{x=1}^{M} \sum_{y=1}^{N} \sqrt{\left(\frac{G_{xn}^2 + G_{yn}^2}{2}\right)} \quad (15)$$

where G_{xn} and G_{yn} are the gradients in the horizontal and vertical direction respectively, and n $\in \{VI_s, IR_s\}$. Then the weights constructed by AVG are calculated as follows:

$$P_{VI_s} = G_{VI_s} / \left(G_{VI_s} + G_{IR_s}\right) \quad (16)$$
$$P_{IR_s} = 1 - P_{VI_s} \quad (17)$$

However, the weights obtained above are misaligned with the boundaries due to the presence of spatial inconsistency and noise in the calculated weights. So we introduce the guided filter to refine the weights obtained by Eqs. (16) and (17), and the refined weights are given as:

$$\omega_{VI_s} = G\left(I_{VI_s}, P_{VI_s}\right) \quad (18)$$
$$\omega_{IR_s} = G\left(I_{IR_s}, P_{IR_s}\right) \quad (19)$$

Finally, the fused saliency part is obtained as

$$F_s = \omega_{VI_s} I_{VI_s}(i,j) + \omega_{IR_s} I_{IR_s}(i,j) \quad (20)$$

The method for the saliency sub-image fusion is summarized in Algorithm2.

Algorithm 2 Fusion of Saliency Sub-Images

Input: Infrared sub-image I_{IR_s}, visible sub-image I_{VI_s}, parameters η, r
Output: Fused saliency part F_s
1. Use the horizontal and vertical gradients of the I_{IR_s} and I_{VI_s} to construct the gradient map by Eq.(15);
2. Get the raw weight map P using gradient map by Eqs.(16) and (17);
3. **Repeat**
4. Refine the raw weight map through guided filter
5. in every window of size $(2r+1)*(2r+1)$;
   Minimize $E(a_k, b_k) = \sum_{i \in v_k} \left((a_k I_i + b_k - P_i)^2 + \eta a_k^2\right)$
6. in every window v_k;
7. **Until** Traverse all the windows in the images;
8. Average results in every v_k to get the refined weights ω_{IR_s} and ω_{VI_s};
9. Calculate the final fused saliency part F_s as $F_s = \omega_{VI_s} I_{VI_s} + \omega_{IR_s} I_{IR_s}$

D. RECONSTRUCTION PROCESS

The final fused image F can be reconstructed by simply adding the fused low-rank part F_{lrr} and the saliency part F_s:

$$F(i,j) = F_{lrr}(i,j) + F_s(i,j) \quad (21)$$

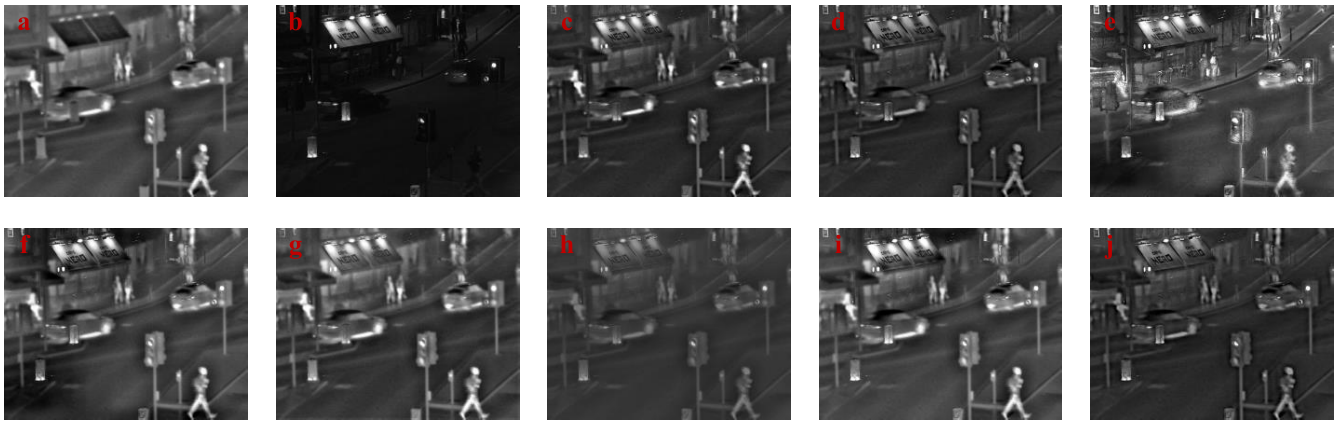


FIGURE 3. Fusion results for image 1 (cars, pedestrians and advertising board). (a) IR; (b) VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.



FIGURE 4. Fusion results for image 2 (plane in the forest). (a) IR; (b) VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. EXPERIMENT SETTING AND IMPLEMENTATION DETAILS

We test the performance of our proposed method on publicly available datasets [21]. The test dataset contains 20 aligned infrared and visible image pairs. Here, we show six pairs of source images from the whole test dataset as examples due to the limited space.

We compare our method with seven popular and classical methods: 1) the co-occurrence filter based method (CoF) [8], 2) the non-subsampled contourlet transform (NSCT) [7], 3) the joint sparse representation based method (JSR) [17], 4) the weighted least square optimization based method (WLS) [14], 5) the infrared feature extraction based fusion (IFE) [27], 6) the multi-resolution singular value decomposition (MSVD) [28] and 7) a convolutional neural networks based method (CNN) [29]. The parameters of all the algorithms are set according to the original papers, and we tuned the programs for the best performance.

The proposed method mainly consist of three parameter types for the LatLRR decomposition, the guided filter and the FGLT-PSO process. In the experiments, we set λ in LatLRR equal to 0.8. The values of parameters of the guided filter are $r = 7$, $\eta = 0.00001$ [9]. The small blur degree we choose here avails to make saliency sub-images spatially smooth. In the FGLT-PSO process part, the setting of the inertia weight ω is critical, because it provides a balance between global and local exploration abilities, which results in a reduction of the optimum solution. According to the experimental results and [30], we set a time decreasing inertia weight value starting from 1.2 and gradually decrease towards 0.4. This is because a large inertia weight at the beginning promotes a global exploration space while the later small inertia weight expedites a fine search. The proper setting of the parameters C_1, C_2 and C_3 in Eq. (12) can result in the alleviation of local minima and a faster convergence procedure. Here, the minimum and maximum default values are the same as used in [22]: $C_{1\,min} = 0.5$, $C_{1\,max} = 2$, $C_{2min} = 1$, $C_{2\,max} = 2$, $C_{3\,min} = 0.5$, $C_{3\,max} = 1.5$. The parameters $rand_1$, $rand_2$ and $rand_3$ are uniformly distributed in the range [0,1].

All experiments are conducted on the Intel Core I7 personal computer, using 3.4GHz with MATLAB codes.

B. SUBJECTIVE PERFORMANCE EVALUATION

Experimental results of the different fusion methods for each of the chosen representative images are shown in Figure 3 – 8. Here, (a) and (b) show the source IR and VI images. The CoF, NSCT, JSR, WLS, IFE, MSVD, CNN



FIGURE 5. Fusion results for image 3. (a) IR; (b) VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.

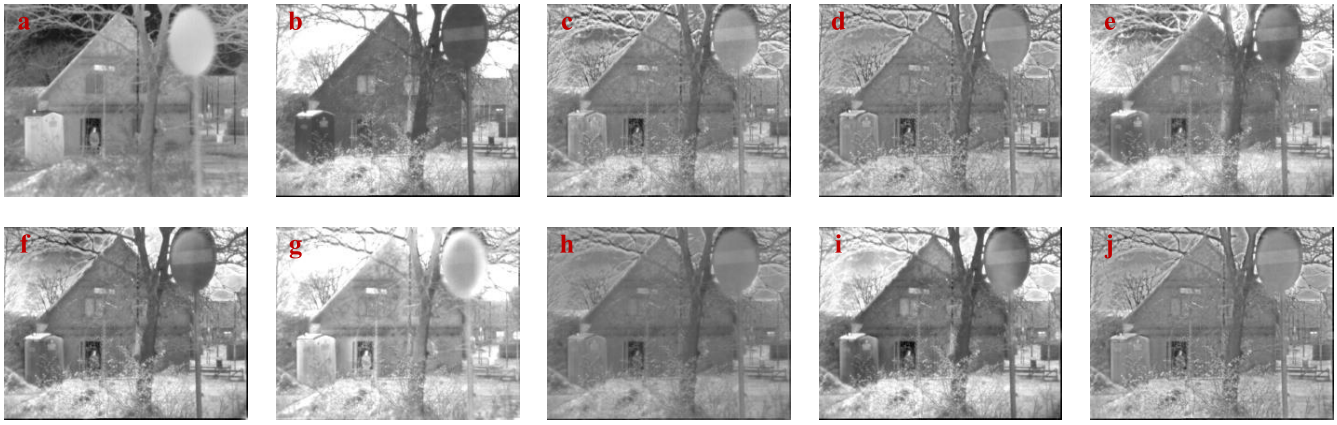


FIGURE 6. Fusion results for image 4. (a) IR; (b) VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.

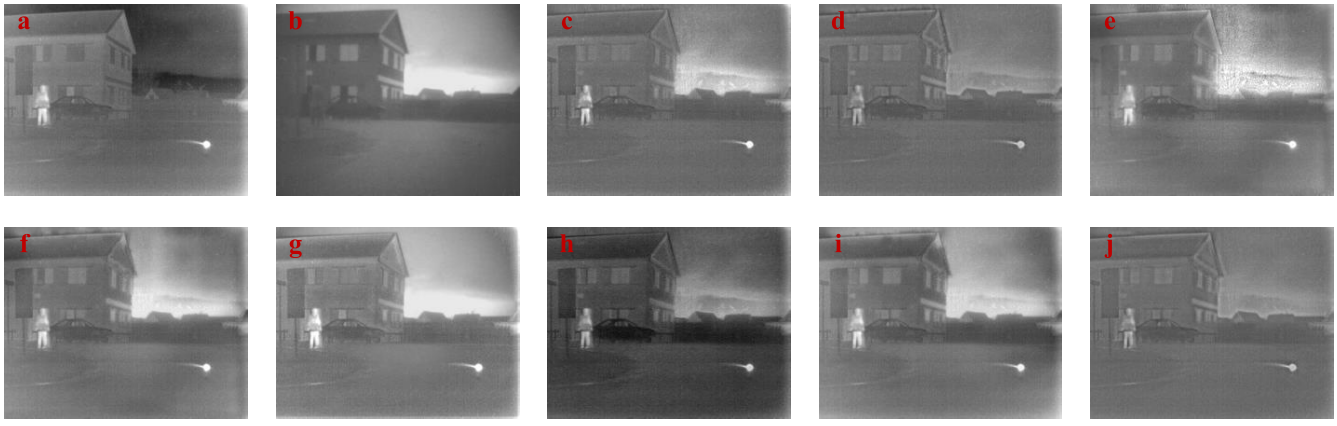


FIGURE 7. Fusion results for image 5. (a) IR; (b)VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.

and the proposed method are shown in succession from (c) to (j).

In Figure 3, cars and pedestrians are clearly seen in the IR image, while the advertising board is clearly visual in the VI image, but not in the IR image. The results obtained from the eight methods are slightly different in contrast and details. The image obtained by our method (j) contains more details of the ''advertising board'' part than the others. In addition, our fusion method contains fewer artifacts and noise compared to the CoF, NSCT and CNN methods. Furthermore, the WLS, JSR, MSVD and IFE methods lack some details and the edges are not very clear.

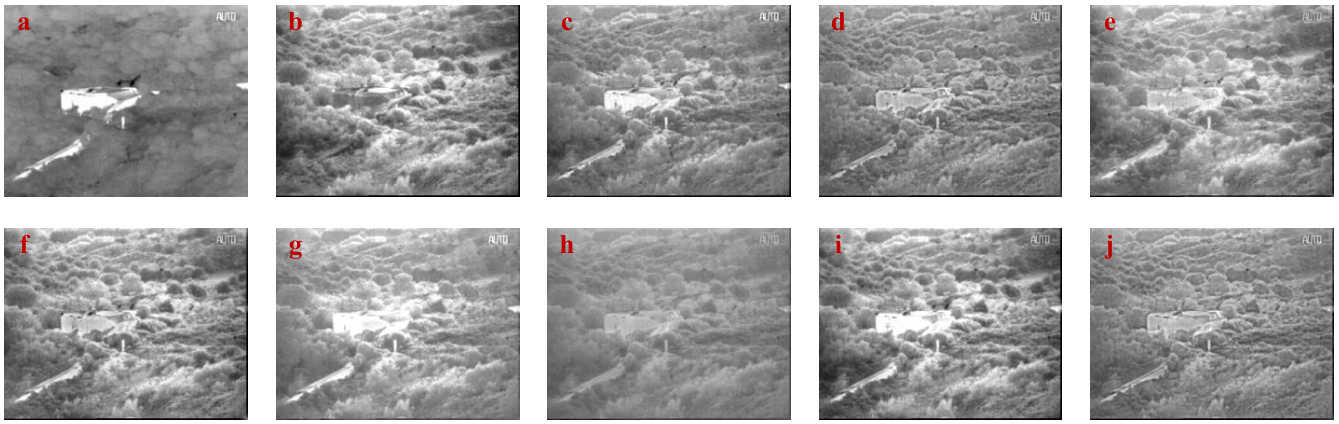


FIGURE 8. Fusion results for image 6. (a) IR; (b) VI; (c) CoF; (d) NSCT; (e) JSR; (f) WLS; (g) IFE; (h) MSVD; (i) CNN; (j) our method.

TABLE 1. Objective evaluation results for Image 1 and 2.

Image	method	SD	$Q^{AB/F}$	$N^{AB/F}$	Q^{SSIM}	Rank score (compared others with our method iterated 10 times)
1	CoF [8]	34.3353	0.5128	0.1115	0.6529	16
	NSCT [7]	31.1957	0.5965	0.1418	0.6595	17
	JSR [17]	31.2194	0.5214	0.1206	0.6445	21
	WLS [14]	35.5376	0.3114	0.4437	0.4978	25
	IFE [27]	34.0297	0.5852	0.2315	0.6709	15
	MSVD [28]	32.4729	**0.6410**	0.1548	0.6536	16
	CNN [29]	32.6730	0.5026	0.1836	0.6305	25
	Proposed(10 times)	35.1203	0.5384	**0.0244**	0.6623	**9**
	Proposed(20 times)	**35.7110**	0.5861	0.0245	**0.6631**	---
2	CoF [8]	45.5382	0.4623	0.1396	0.8134	20
	NSCT [7]	42.5461	0.5671	0.1229	0.8418	13
	JSR [17]	42.1356	0.4328	0.1248	0.8269	22
	WLS [14]	44.7359	0.4660	0.1669	0.8174	20
	IFE [27]	41.7668	0.4046	0.1734	0.7969	32
	MSVD [28]	44.6550	**0.5868**	0.1269	0.8370	13
	CNN [29]	**45.7765**	0.4870	0.1032	0.8017	13
	Proposed(10 times)	44.1270	0.4762	0.0275	0.8637	**11**
	Proposed(20 times)	44.8006	0.4896	**0.0275**	**0.8637**	---

Figure 4 shows an airplane in a forest. One can clearly see that NSCT and WLS produce obvious artifacts in the background outline. The CNN method does not show satisfactory contrast. The JSR method does not contain the most information about the background. The MSVD method does not present the textures of the trees. Furthermore, the IFE method does not perform well in the terms of contrast. Both CoF and the proposed method preserve the details well, but our method also suppresses artifacts to some extent.

As shown in Figure 5, our method contains more details at the window part and more textures of the grass. In addition, the fused image has a higher contrast and less noise.

One can see clearly in the Figure 6 that our method can preserve the texture information of the branches of the

TABLE 2. Objective evaluation results for Image 3 and 4.

Image	method	SD	$Q^{AB/F}$	$N^{AB/F}$	Q^{SSIM}	Rank score (compared others with our method iterated 10 times)
3	CoF [8]	16.3752	0.4821	0.2319	0.6525	21
	NSCT [7]	19.4820	0.6084	0.1358	0.6342	13
	JSR [17]	15.6920	0.4167	0.1412	**0.7247**	19
	WLS [14]	14.3324	0.3725	0.2468	0.6132	31
	IFE [27]	16.3277	0.3206	0.0274	0.6657	20
	MSVD [28]	**20.1643**	**0.6487**	0.1384	0.6275	13
	CNN [29]	16.7850	0.4836	0.2195	0.6821	16
	Proposed(10 times)	16.7921	0.4764	0.0044	0.6895	**11**
	Proposed(20 times)	16.9782	0.4842	**0.0044**	0.6901	---
4	CoF [8]	44.2760	0.4589	0.1374	0.7368	14
	NSCT [7]	42.4895	0.5302	0.1251	**0.7729**	12
	JSR [17]	41.4692	0.4429	0.1216	0.6771	25
	WLS [14]	43.2180	0.4572	0.1775	0.6520	27
	IFE [27]	**48.3750**	0.4538	0.1675	0.7293	20
	MSVD [28]	41.5643	**0.5561**	0.1366	0.7322	16
	CNN [29]	45.1693	0.4585	0.1580	0.7256	18
	Proposed(10 times)	43.9318	0.4579	**0.0128**	0.7441	**12**
	Proposed(20 times)	44.4356	0.4593	0.0129	0.7429	---

background and the chairs in the right part, and highlight the person next to the door at the same time.

Similar to the previous results, in Figure 7 and 8, the results of the proposed method is superior to the other methods in terms of visual effect.

To sum up, our method can eliminate the artifacts and noise produced by spatial inconsistency due to the effect of the guided filter and contains more saliency features and more detail information. What is more, the proposed method has a good and stable performance for different kinds of images.

C. OBJECTIVE PERFORMANCE EVALUATION

Generally, one can get an intuitive impression of the performance of the algorithms through subjective evaluations of the fused images. However, it is difficult to compare the fusion quality accurately and without bias only by subjective assessments. Likewise, a single evaluation metric cannot measure the quality of images comprehensively. Therefore, in order to quantitatively assess the performance of the different fusion methods, the four indices briefly introduced as follows are adopted as the evaluation criteria: 1) edge retentiveness ($Q^{AB/F}$) [24], 2) standard deviation (SD) [25], 3) the image structure-based metric (Q^{SSIM}) [31] and 4) fusion artifacts ($N^{AB/F}$) [32]. The image feature-based metric $Q^{AB/F}$ measures how well the amount of edge information is kept in the fused images. The metric SD is commonly used to measure the overall contrast of the gray values. The image structure-based metric Q^{SSIM} uses structural similarity (SSIM) to assess the amount of structural similarity transferred from the source images to the fused images. The metric $N^{AB/F}$ reflects the rate of noise and artifacts introduced by a fusion process into the fused image. Higher values of $Q^{AB/F}$, SD and Q^{SSIM} indicate a better fusion quality of the fused image. In contrast, when the value of $N^{AB/F}$ is small, the fusion performance is better.

We need to compare the different algorithms combined with the various evaluation metrics comprehensively. However, diverse methods perform differently in various criteria and we cannot analyze performance in each metric in isolation. Therefore, we set up a scoring mechanism called rank score to compare the eight methods taking all the metrics into consideration. The rule is as follows: Assuming that all the metrics are equal important (although in specific applications, different weights can be set for various criteria to meet different needs), we count the ranking of the eight methods under each evaluation metric in each image pair and record 1-8 according to the ranking order of the eight algorithms:

$$\begin{aligned}\text{Rank} = {} & \text{Rank(SD)} + \text{Rank}(Q^{AB/F}) + \text{Rank}(Q^{SSIM}) \\ & + \text{Rank}(N^{AB/F}) \end{aligned} \tag{22}$$

TABLE 3. Objective evaluation results for Image 5 and 6.

Image	method	SD	$Q^{AB/F}$	$N^{AB/F}$	Q^{SSIM}	Rank score (compared others with our method iterated 10 times)
5	CoF [8]	31.3296	0.4923	0.1868	0.6892	22
	NSCT [7]	**37.6597**	0.5620	0.1414	0.6866	13
	JSR [17]	32.1152	0.4901	0.1501	0.7138	20
	WLS [14]	32.6765	0.4020	0.2512	0.7209	22
	IFE [27]	36.7853	0.3244	0.1059	0.6657	20
	MSVD [28]	34.3950	**0.5876**	0.1395	0.6794	14
	CNN [29]	29.4736	0.4577	0.1955	0.7182	24
	Proposed(10 times)	33.4720	0.4944	0.0092	0.7234	**9**
	Proposed(20 times)	34.4168	0.5002	**0.0092**	**0.7276**	---
6	CoF [8]	14.1930	0.4572	0.1502	0.7962	20
	NSCT [7]	15.8194	0.5388	0.1136	0.8035	10
	JSR [17]	12.7725	0.4375	0.0832	0.6260	24
	WLS [14]	13.731	0.4568	0.1127	0.6242	22
	IFE [27]	**16.3078**	0.4551	0.2220	0.7554	21
	MSVD [28]	12.9429	**0.5404**	0.1457	0.7849	18
	CNN [29]	14.2172	0.5173	0.1328	0.6226	20
	Proposed(10 times)	14.2303	0.4860	**0.0060**	0.8350	**9**
	Proposed(20 times)	14.7160	0.5180	0.0061	**0.8358**	---

Smaller values of the rank score indicate a better overall performance of the algorithm. For instance, in Table 1, image1, the CoF method ranks 3 in SD, 6 in $Q^{AB/F}$, 2 in $N^{AB/F}$ and 5 in Q^{SSIM}, so the final rank score is 16 $(3 + 6 + 2 + 5)$. In the same way, the final rank score of CNN is 25, which shows that the performance of CoF is better than CNN according to the overall evaluation metrics.

Quantitative comparisons of the performance among the different methods are shown in the Tables 1-3 for the six pairs of sample images. The last two lines in each example are the results of the proposed method for 10 iterations and 20 iterations. Values in boldface represent the best results with the same index.

The rank scores are shown in the last column. Note that we only compared the rank score of the other seven methods and our method iterated 10 times. The results illustrate that our algorithm makes better trade-off between efficiency and quality.

We can observe from Table 1-3 that for the proposed method (20 iterations), there are two almost best values (Q^{SSIM} and $N^{AB/F}$) in the four evaluation indexes, while the $Q^{AB/F}$ and SD measures are in the top three overall. This indicates that the proposed method can provide more saliency information and preserve more detailed information compared to other fusion methods. In addition, the proposed method produces less artifacts and noise, which is very crucial for the visual impact and subsequent image processing. When comparing our method (10 iterations) with other seven methods, even though the scores of proposed method are not the best in some criteria, the rank score we defined of the proposed method are always the best, which indicates that our method is superior to the other methods in terms of adaptivity and robustness for different types of source images. Besides, our method can make a good compromise between efficiency and quality by setting the number of iterations.

The complexity and efficiency of the different algorithms are evaluated by the running time. We calculated the running time of 20 pairs of images and the final running time shown in Table 4 is obtained by the average of the 20 pairs of images with 8 times operations. All experiments are conducted on the same condition. The CNN and JSR cost too much time due to their own operating scheme and they are not real-time. The running time of another six methods are real-time and may be a little different with different decomposition layers and parameters. Although during the low-rank parts fusion in our method, applying 20 iterations more than doubles the time

TABLE 4. Running time of different image fusion methods (seconds).

Methods	CoF	NSCT	JSR	WLS	IFE	MSVD	CNN	Proposed(10's)	Proposed(20's)
Running Time	5.6239	4.7986	47.7261	0.9339	2.7462	0.6273	76.4775	4.2967	9.7342

of 10 iterations, the running time to make twenty iterations is acceptable and can be applied in practice. To sum up, our method is not the fastest and the running time increases as the iteration times increase, it is still efficient compromising the running time and the quality.

V. CONCLUSION

In this paper, we present an adaptive two-scale image fusion scheme based on guided filter and fusion Global-Local-Topology Particle Swarm Optimization. We adopt LatLRR to generate two-scale sub-images and reconstruct the fused images by adding the fused low-rank parts and the saliency parts, which can reduce noise and the ringing phenomenon effectively. FGLT-PSO is using to construct the adaptive weights to make the fusion performance stable and robust. Meanwhile, guided filter can suppress artifacts to improve the clarity of the saliency parts. In addition, our method is simple and easy to operate: only adjust a few parameters will balance the quality and efficiency of the algorithm. Relevant experiments are conducted to evaluate the performance of the proposed method. The experiments show that the proposed method outperforms other fusion methods in terms of both visual quality and objective evaluation.

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